Evaluating (11) at $x = R$ and using (14) to substitute $S(\cdot)$ yields what is usually referred to as the job-destruction condition:

$$R - \phi - c - \left(\frac{b}{k} + \frac{\beta}{1-\beta}c\theta\right) + \frac{\lambda}{r+\delta+\lambda}\int_R (x-R)dG(x) = 0. \tag{15}$$

As is standard, the destruction decision is independent of scale if b is. The natural interpretation of b is that it is unemployment insurance income. Along these lines, if one lets $b = \tau_b E_G[w(x) \mid x \geq R]$, where $\tau_b \in [0,1)$ is the replacement rate, then $b = \hat{b}k$, with

$$\hat{b} = \frac{\tau_b \beta[\tilde{x}(R) - \phi - c + c\theta]}{1-(1-\beta)\tau_b},$$

and $\tilde{x}(R) \equiv E_G[x \mid x \geq R] = [1-G(R)]^{-1}\int_R x dG(x)$. Under this specification, b is linear in k so (15) is independent of k and becomes

$$R - \frac{\tau_b \beta \tilde{x}(R)}{1-(1-\beta)\tau_b} - \frac{(1-\tau_b)(\phi+c)}{1-(1-\beta)\tau_b} - \frac{\beta c\theta}{(1-\beta)[1-(1-\beta)\tau_b]} + \frac{\lambda}{r+\delta+\lambda}\int_R (x-R)dG(x) = 0.$$

In what follows I will always abstract from scale effects caused by unemployment income b by assuming it is a fraction of the average going wage.

Substituting the equilibrium condition $rV = 0$ in (6) implies $(1-\beta)\int_R S(x)dG(x) = \frac{ck}{q(\theta)}$. That is, the expected profit from a filled job equals the expected hiring cost in an equilibrium with free entry. Using (14) to substitute $S(\cdot)$ out of this expression yields what is often referred to as the job-creation condition:

$$\frac{1-\beta}{r+\delta+\lambda}\int_R (x-R)dG(x) - \frac{c}{q(\theta)} = 0. \tag{16}$$

The job-creation and job-destruction conditions jointly determine R and θ, and under the maintained assumptions, they are independent of scale, k.[17] For given c and ϕ, an equilibrium is a vector $[\theta, R, H, U, w, u, K]$ such that (θ, R) jointly solve (15) and (16); given (θ, R), H satisfies (3); U is given by (10); w is given by (12); and u is given by (2). In addition, the market for capital services should clear, so the aggregate supply of capital K must satisfy $K = [1-(1-\theta)u]k$, where the R.H.S. is the total demand for capital (coming from both matched and unmatched firms). Note that in parametrizations that result in $R < \phi$, the capital and workers in matches with realizations in $[R, \phi)$ remain employed but are not engaged in production. The firms in these states have excess capacity and hoard labour. The following section provides a sharper characterization of aggregate outcomes for a particular distribution of idiosyncratic shocks.

4. AGGREGATION

Let K_e denote the capital in place at all the firms with filled jobs, that is,

$$K_e = \frac{1-u}{1-(1-\theta)u}K. \tag{17}$$

17. See Lemma 1 in Appendix B for conditions under which the pair (θ, R) that solves (15) and (16) exists and is unique.

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Aggregate output, Y, and the total number of hours worked, N, are given by

$$Y = (1-u)\int_{\mu} f[x, n(x), k]dH(x),$$

and $N = (1-u)\int_{\mu} n(x)dH(x)$, with $\mu \equiv \max(R, \phi)$. Using (1) and (8) gives

$$Y(K_e, \mu) = [1 - H(\mu)]K_e E_H(x \mid x \geq \mu) \tag{18}$$

$$N = [1 - H(\mu)]K_e, \tag{19}$$

where $E_H(x \mid x \geq \mu) = [1 - H(\mu)]^{-1}\int_{\mu} xdH(x)$. Intuitively, since every firm–worker pair is setting hours either to 0 or to full capacity k, the aggregate number of hours worked is just equal to the fraction of firm–worker pairs who engage in production times the total capital stock in filled jobs. Similarly, aggregate output equals the number of *active* units of capital, $[1 - H(\mu)]K_e$, weighted by their average productivity.[18] Following Houthakker (1955–1956), one could imagine solving (19) for the "aggregate labour demand" by active firms, $\mu(K_e, N)$, and then substituting it in (18) to obtain $Y[K_e, \mu(K_e, N)]$. Hereafter, I use $F(K_e, N)$ to denote $Y[K_e, \mu(K_e, N)]$ to simplify notation and to stress the fact that this is the economy's *aggregate production function.* This aggregate production function exhibits constant returns to scale even for an arbitrary H. To see this, notice that $\mu(K_e, N)$ is homogeneous of degree 0, and hence (18) indicates that for any $\zeta > 0$, $F(\zeta K_e, \zeta N) = \zeta F(K_e, N)$.[19]

Suppose idiosyncratic shocks are draws from a Pareto distribution

$$G(x) = \begin{cases} 0 & \text{if } x < \varepsilon \\ 1 - \left(\frac{\varepsilon}{x}\right)^{\alpha} & \text{if } \varepsilon \leq x, \end{cases} \tag{20}$$

where $\varepsilon > 0$ and $\alpha > 1$. Then, provided $R \geq \varepsilon$, $1 - G(R) = \left(\frac{\varepsilon}{R}\right)^{\alpha}$, and hence $G(x) - G(R) = (\varepsilon/R)^{\alpha}[1 - (R/x)^{\alpha}]$ for any $x \geq R$. Substituting these expressions in (3) yields

$$H(x) = \begin{cases} 0 & \text{if } x < R \\ 1 - \left(\frac{R}{x}\right)^{\alpha} & \text{if } R \leq x. \end{cases} \tag{21}$$

So the steady-state productivity distribution of active matches follows a Pareto distribution with parameters R and α. Using (21), $1 - H(\mu) = \left(\frac{R}{\mu}\right)^{\alpha}$, and $E_H(x \mid x \geq \mu) = \frac{\alpha}{\alpha-1}\mu$, so the aggregates (18) and (19) specialize to $Y(K_e, \mu) = \frac{\alpha}{\alpha-1}R^{\alpha}\mu^{1-\alpha}K_e$, and $N = (R/\mu)^{\alpha}K_e$. Inverting the latter yields $\mu = (K_e/N)^{1/\alpha}R$, and substituting this in the former implies

$$F(K_e, N) = AK_e^{\gamma}N^{1-\gamma}, \tag{22}$$

18. As mentioned in footnote 8, Mortensen and Pissarides (1994) assume that G has support [0, 1] and that all new matches start off with productivity 1. So, with $\delta = 0$, aggregate output in their model evolves according to

$$\dot{Y} = \theta q(\theta)uk - \lambda Y + \lambda(1-u)k\int_{\mu}^{1} xdG(x).$$

Replacing $(1-u)k$ with K_e, steady-state output is $Y = \frac{\theta q(\theta)uk}{\lambda} + [1 - H(\mu)]K_e E_H(x \mid x \geq \mu)$, which looks like (18) except for the first term. Assuming that the initial productivity of a new match is a random draw from G—just as the innovations to the productivity of ongoing matches—allows for a density G with unbounded support. In addition, this alternative assumption smooths aggregate output by getting rid of the "spike" $\theta q(\theta)uk\lambda^{-1}$.

19. Also, from (18), $F_2(K_e, N) = -\mu_2(K_e, N)K_e\mu dH(\mu)$, and from (19), $-\mu_2(K_e, N)K_e dH(\mu) = 1$. Thus, $F_2(K_e, N) = \mu$. So the marginal product of labour in the aggregate production function is equal to the marginal product of the least efficient unit of labour employed in production. I owe this argument to Erzo G. J. Luttmer.

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where

$$A = \frac{R}{1-\gamma} \tag{23}$$

and $\gamma \equiv 1/\alpha$. This aggregation result is reminiscent of the classic aggregation result of Houthakker (1955–1956).[20] The factor A is what macroeconomists normally refer to as TFP. Its level depends on α, a parameter of the primitive distribution of productivity shocks, as well as on all the characteristics of the labour market as summarized by the destruction decision, R.

5. LABOUR-MARKET POLICIES AND THE LEVEL OF TFP

This section considers the effects of four policies: employment subsidies, hiring subsidies, firing taxes, and unemployment benefits. I follow Pissarides (2000) and model the subsidies as transfers from the government to the firm and the firing taxes as payments from the firm to the government.[21] The value function $W(x)$ is still given by (5), while (4), (6), and (7) generalize to

$$rU = \tau_b + \theta q(\theta) \int \max[W_o(z) - U, 0] dG(z)$$

$$rV = -ck + q(\theta) \int \max[J_o(z) + \tau_h k - V, 0] dG(z)$$

$$rJ(x) = \pi(x) + \tau_e k + \lambda \int \max[J(z) - V + \tau_f k, 0] dG(z) - (\delta + \lambda)[J(x) - V + \tau_f k].$$

The policy variables are a hiring subsidy, τ_h, an employment subsidy, τ_e, a firing tax, τ_f, and unemployment benefits, τ_b.[22] In Appendix B I show that if G is as in (20), the job-destruction and job-creation conditions that determine the equilibrium pair (θ, R) are

$$R - \phi - c + \tau_e + r\tau_f - \left(\tau_b + \frac{\beta}{1-\beta} c\theta\right) + \frac{\lambda \varepsilon^\alpha R^{1-\alpha}}{(\alpha - 1)(r + \delta + \lambda)} = 0 \tag{24}$$

$$\frac{\varepsilon^\alpha R^{1-\alpha}}{(\alpha - 1)(r + \delta + \lambda)} + \left(\frac{\varepsilon}{R}\right)^\alpha (\tau_h - \tau_f) - \frac{c}{(1-\beta) q(\theta)} = 0. \tag{25}$$

The main properties of the equilibrium are summarized as follows.

Proposition 1. *Suppose* $\varepsilon + (\alpha - 1)(r + \delta + \lambda)(\tau_h - \tau_f) > 0$. *Let* θ_ϵ^* *be defined by* $q(\theta_\epsilon^*) = \frac{(\alpha-1)(r+\delta+\lambda)c}{(1-\beta)[\varepsilon+(\alpha-1)(r+\delta+\lambda)(\tau_h-\tau_f)]}$, *and* $\phi_\epsilon = \left[1 + \frac{\lambda}{(\alpha-1)(r+\delta+\lambda)}\right]\varepsilon - \left(1 + \frac{\beta}{1-\beta}\theta_\epsilon^*\right)c + \tau_h + r\tau_f - \tau_b$. *If* $\varepsilon + \alpha(r + \delta + \lambda)(\tau_h - \tau_f) > 0$, *then for any* $\phi > \phi_\varepsilon$ (a) *there exists a unique equilibrium,* (b) $R > \varepsilon$, (c) $\partial R/\partial\phi > 0$, *and* (d) $\partial\theta/\partial\phi < 0$. *If in addition* $\phi_\epsilon - \varepsilon > 0$, *then* (e) *there is a nondegenerate interval* $(\phi_\epsilon, \widetilde{\phi})$ *such that* $R(\phi) < \phi$ *for all* $\phi \in (\phi_\epsilon, \widetilde{\phi})$.

Aggregate output is still given by (22); the aggregate stock of capital in filled jobs, K_e, is still given by (17); and the aggregate number of hours worked, N, is still given by $N = (R/\mu)^\alpha K_e$. In addition, if the measure of capital used to construct aggregate output is K_e, then the level of TFP

20. I discuss this relationship at some length in Section 6.

21. I assume that upon separation, the firm must pay the firing tax to the government because in the present set-up, firing taxes would be completely neutral under the alternative scheme in which the firm compensates the fired worker directly.

22. All payments are assumed to be proportional to the firm's size, as measured by k. For example, upon separation, the firm is taxed $\tau_f k$, and while unemployed, the worker receives $\tau_b k$ from the government. This assumption, borrowed from Pissarides (2000), is convenient because it ensures that policies introduce no scale effects into the job-creation and job-destruction decisions. Also, to keep the analysis simple, I will ignore the government's financing constraints. A natural extension would be requiring the government to run a balanced budget. A simple example of a scheme that is self-financing in the steady state is $\tau_f = \tau_h$, and $\tau_b = \tau_e = 0$.

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